

Assignment 2(a)

Title: Write a program to implement Parallel Bubble Sort using OpenMP

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```
#include <iostream>
```

```
#include <omp.h>
```

```
using namespace std;
```

```
/* Swap function */
```

```
void swapValues(int &a, int &b)
```

```
{
```

```
    int temp = a;
```

```
    a = b;
```

```
    b = temp;
```

```
}
```

```
/* Parallel Bubble Sort (Odd-Even Transposition Sort) */
```

```
void bubble(int *a, int n)
```

```
{
```

```
    for (int i = 0; i < n; i++)
```

```
    {
```

```
        /* Even phase */
```

```
        #pragma omp parallel for
```

```
        for (int j = 0; j < n - 1; j += 2)
```

```
        {
```

```
            if (a[j] > a[j + 1])
```

```
                swapValues(a[j], a[j + 1]);
```

```
        }
```

```
        /* Odd phase */
```

```
        #pragma omp parallel for
```

```
        for (int j = 1; j < n - 1; j += 2)
```

```
        {
```

```
            if (a[j] > a[j + 1])
```

```
                swapValues(a[j], a[j + 1]);
```

```
        }
```

```
    }
```

```
}
```

```
int main()
```

```
{
```

```
int n;

cout << "Enter total number of elements: ";
cin >> n;

int *a = new int[n];

cout << "Enter elements:\n";
for (int i = 0; i < n; i++)
    cin >> a[i];

bubble(a, n);

cout << "\nSorted array:\n";
for (int i = 0; i < n; i++)
    cout << a[i] << " ";

cout << endl;

delete[] a;
return 0;
}
```

OUTPUT:

```
datanalytics1@datanalytics1-OptiPlex-7050:~$ nano parallel_bubble.cpp
datanalytics1@datanalytics1-OptiPlex-7050:~$ g++ -fopenmp parallel_bubble.cpp -
o parallel_bubble
datanalytics1@datanalytics1-OptiPlex-7050:~$ ./parallel_bubble
Enter total number of elements: 6
Enter elements:
9 3 7 1 5 2

Sorted array:
1 2 3 5 7 9
datanalytics1@datanalytics1-OptiPlex-7050:~$
```